

Anomaly Detection in Medicare Part D Opioid Prescribing
A Reproducible Dual-Method Risk Stratification Framework with Temporal Holdout Validation

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Working Paper Version 1.0

Prepared 2026-05-13 for submission to SSRN Health Economics Network

Abstract

Background. The HHS Office of Inspector General has identified opioid prescribing anomalies as a Medicare program integrity priority. Existing federal capability for prospective identification of statistical billing outliers among Medicare Part D opioid prescribers is limited and fragmented.

Methods. We construct a reproducible anomaly detection and risk stratification framework on the CMS Medicare Part D Prescriber by Provider public use file for calendar year 2023, encompassing 1,380,665 prescribers. After analytical universe filtering and explicit clinical exclusions for oncology and palliative specialties, the modeling universe comprises 1,025,487 prescribers across 191 specialties and 61 jurisdictions. We engineer six per prescriber risk indicator features, compute a peer state plus specialty weighted z-score statistical risk layer, train an Isolation Forest machine learning anomaly detector at contamination 0.05 with random seed 42 on a standardized eight feature matrix, and combine both layers into a 0 to 100 anomaly score with the top 5 percent flagged for investigative prioritization. We validate findings with six pillars: bootstrap confidence intervals (1,000 iterations), permutation feature importance, temporal holdout validation against 2019 through 2022 training pool with 2023 holdout, OIG List of Excluded Individuals and Entities concordance, DOJ defendant cross reference, and OIG controlled substance fraud typology alignment.

Results. The framework identifies 51,275 flagged prescribers (5.0 percent of the modeling universe), with directly observed sample excess of \$1.16B above peer state plus specialty median prescribing. Bootstrap 95 percent confidence interval is \$1.14B to \$1.18B, with relative standard deviation 0.78 percent. Coverage adjusted projected national excess is \$4.22B using a 27.5 percent Medicare opioid prescription share multiplier sourced from CDC and SAMHSA data. Method concordance between Isolation Forest and combined score is 92.0 percent. Temporal holdout coverage from 2019 through 2021 training pool to 2023 holdout is 70.5 percent for prescribers flagged in two or more training years. OIG LEIE direct NPI concordance is 29 matches.

Conclusions. The framework provides an audit grade methodology for prospective Medicare Part D opioid anomaly detection that is fully reproducible from public federal sources. Results are robust under bootstrap resampling and demonstrate substantial temporal persistence across years. The framework is suitable for adoption by federal program integrity agencies for investigative prioritization, with explicit acknowledgment that anomaly scores are not fraud probabilities and that individual review by appropriate authorities is required before any inference.

Keywords

Medicare Part D; opioid prescribing; anomaly detection; isolation forest; program integrity; healthcare fraud surveillance; peer benchmarking; bootstrap; temporal validation; OIG.

JEL Classification

I11 (Analysis of Health Care Markets), I13 (Health Insurance, Public and Private), I18 (Government Policy, Regulation, Public Health), H51 (Government Expenditures and Health), C45 (Neural Networks and Related Topics), C53 (Forecasting and Prediction Methods).

1. Introduction

Medicare Part D, the prescription drug component of Medicare, processes more than four hundred billion dollars in annual drug spending. The CMS Fiscal Year 2025 Improper Payments Fact Sheet reports a Part D improper payment rate of 4.00 percent representing \$4.23 billion. Opioid prescribing has been a sustained federal program integrity priority. The HHS Office of Inspector General publishes recurring audits and semiannual reports identifying anomalous opioid prescribing as an area requiring continued oversight. The Department of Justice announced a **\$300 million to \$350 million Walgreens Boots Alliance opioid settlement** in 2024 evidencing continued federal enforcement on opioid distribution. The MedPAC March 2025 Report to Congress and the HHS-OIG Fall 2024 Semiannual Report both identify Medicare Part D opioid prescribing as an ongoing area of audit focus.

The SUPPORT for Patients and Communities Act, Public Law 115-271, Section 6052, authorizes the Centers for Medicare and Medicaid Services to award grants and contracts to educate and conduct outreach to outlier opioid prescribers and to reduce the volume of opioids they prescribe; this work identifies the outlier prescribers that such programs are designed to target. GAO 23-105632 documents prescription opioid use, spending, and trends in Medicare Part D. CDC US Prescribing Rate Maps document the geographic distribution of opioid prescribing across all payers.

This paper presents a reproducible anomaly detection and risk stratification framework for Medicare Part D opioid prescribing. The contribution is fivefold. First, the framework is **fully reproducible** from public federal sources, with source file SHA-256 hashes, software environment versions, and explicit hyperparameters captured in a reproducibility envelope. Second, the framework uses **dual-method risk scoring** combining a weighted z-score statistical layer with an Isolation Forest machine learning layer, with method concordance reported transparently. Third, the framework includes a **six pillar validation suite** covering bootstrap confidence intervals, permutation feature importance, temporal holdout, OIG concordance, DOJ cross reference, and pattern alignment. Fourth, the framework reports both **sample and projected national figures separately**, with the projection methodology fully documented and the multiplier sourced from CDC and SAMHSA payer mix data. Fifth, the framework is **explicitly framed as anomaly detection, not fraud detection**: risk scores are anomaly scores measuring statistical deviation from peer billing patterns, not fraud probabilities; flagged prescribers warrant investigative prioritization by appropriate authorities, not enforcement determinations.

2. Methods

2.1 Data Sources

Five public federal data sources are used. The **CMS Medicare Part D Prescriber by Provider** public use files for calendar years 2019 through 2023 form the primary analytical input. The **HHS-OIG List of Excluded Individuals and Entities** provides the OIG concordance reference set. The **CMS Medicare Part D Opioid Prescribing Rates by Geography** file for 2023 provides state level baseline rates. The **CDC National Vital Statistics System Provisional Drug Overdose Death Counts** provide state level mortality reference data. The **CMS Medicare Part D Spending by Drug** file for 2023 provides drug level spending context.

2.2 Analytical Universe

The 2023 CMS Medicare Part D Prescriber by Provider file contains **1,380,665 prescriber records**. Application of the analytical universe filter (non null opioid total claims) isolates **1,033,188 opioid prescribers**. Clinical exclusions for **7,701 oncology and palliative specialty providers** produce a modeling

universe of **1,025,487 prescribers** operating across **191 specialties** and **61 jurisdictions**. The clinical exclusion list is explicit: Hematology Oncology, Medical Oncology, Radiation Oncology, Gynecological Oncology, Surgical Oncology, Hospice and Palliative Care, Hospice and Palliative Medicine.

2.3 Feature Engineering

Six per prescriber risk indicator features are engineered with explicit formulas documented in the result file `feature_definitions.csv`.

1. **Opioid intensity ratio:** `opiod_tot_clms` divided by `opiod_tot_benes`. Per beneficiary opioid claim intensity.
2. **Cost per beneficiary:** `opiod_tot_drug_cst` divided by `opiod_tot_benes`. Per beneficiary opioid drug cost.
3. **Average days supply:** `opiod_tot_suply` divided by `opiod_tot_clms`. Average days supply per claim.
4. **Long acting opioid share:** 100 times `opiod_la_tot_clms` divided by `opiod_tot_clms`. Long acting opioid share of all opioid claims.
5. **Peer deviation:** (`opiod_tot_clms` minus `state_specialty_median`) divided by (`state_specialty_median` plus 1). Deviation from peer state plus specialty median.
6. **Volume percentile:** Percentile rank of `opiod_tot_clms` within the modeling universe.

2.4 Peer Baselines

Peer baselines are computed at two granularities. State plus specialty cohorts with minimum 30 prescribers (n equals 2,020 cohorts) provide the primary benchmark. National baselines provide a fallback for cohorts below the minimum size. Robust statistics (median) are used for peer deviation; mean and standard deviation are used for the statistical risk z-score layer.

2.5 Statistical Risk Layer

Per feature absolute z-scores against national mean and standard deviation are weighted as follows: opioid intensity ratio 30 percent, cost per beneficiary 20 percent, peer deviation 15 percent, average days supply 15 percent, long acting share 10 percent, volume percentile 10 percent. Weights sum to 100 percent. The weighted absolute z-score sum is the statistical risk raw score.

2.6 Machine Learning Risk Layer

An Isolation Forest is trained on an eight feature matrix comprising the six engineered features plus `opiod_tot_clms` and `opiod_prscrbr_rate` as raw scale features. Parameters are explicit: `contamination` 0.05, `n_estimators` 100, `random_state` 42, `max_samples` 256, `bootstrap` false. `StandardScaler` standardization is applied before fitting. Decision function output is inverted so higher values correspond to greater anomaly.

2.7 Combined Score and Tier Stratification

Both layers are min max normalized to 0 to 1 and combined at 60 percent statistical and 40 percent ML weights. The result is rescaled to 0 to 100. Percentile based tier cutoffs assign Low (under 80th), Medium (80th to 95th), High (95th to 99th), Critical (99th to 99.9th), and Extreme (99.9th and above) tiers. The flagged set is the top 5 percent of the modeling universe.

2.8 Validation Procedures

Bootstrap. The modeling universe is resampled with replacement 1,000 times using fixed random seed 42. On each resample, the flagged threshold is recomputed at the 95th percentile and sample excess is summed. The 2.5th and 97.5th percentiles bound the 95 percent confidence interval.

Permutation importance. For each of the eight ML features, the feature column is shuffled within a 100,000 row stratified sample and the Spearman rank correlation drop between the resulting and original Isolation Forest scores is averaged across five repeats. Normalized to sum to 100 percent.

Temporal holdout. The pipeline is run on each of 2019, 2020, 2021 to produce per year flagged NPI sets. NPIs flagged in two or more of the three training years are considered persistent. Holdout coverage is the fraction of persistent NPIs that appear in the 2023 holdout universe and are independently flagged in 2023.

OIG concordance. Direct NPI matching of the flagged set against the OIG LEIE, with NPI equals zero placeholders filtered, produces direct match count. Bucketed fuzzy name matching for top 20 flagged at SequenceMatcher similarity threshold 0.85 produces additional matches.

DOJ cross reference. Thirteen documented opioid prosecution defendants are matched against the active prescriber file via fuzzy comparison with token bucketing at similarity threshold 0.85.

Pattern alignment. Five OIG controlled substance fraud typologies with explicit threshold definitions are computed: extreme prescription concentration, high dose extended supply, long acting opioid concentration, excess cost per beneficiary, and multi state operation indicator.

3. Results

3.1 Headline Findings

The Isolation Forest at contamination 0.05 identifies **51,275 anomalies** on the modeling universe. The combined risk score top 5 percent flag is **51,275 prescribers**. Method concordance is **92.0 percent**.

3.2 Risk Tier Distribution

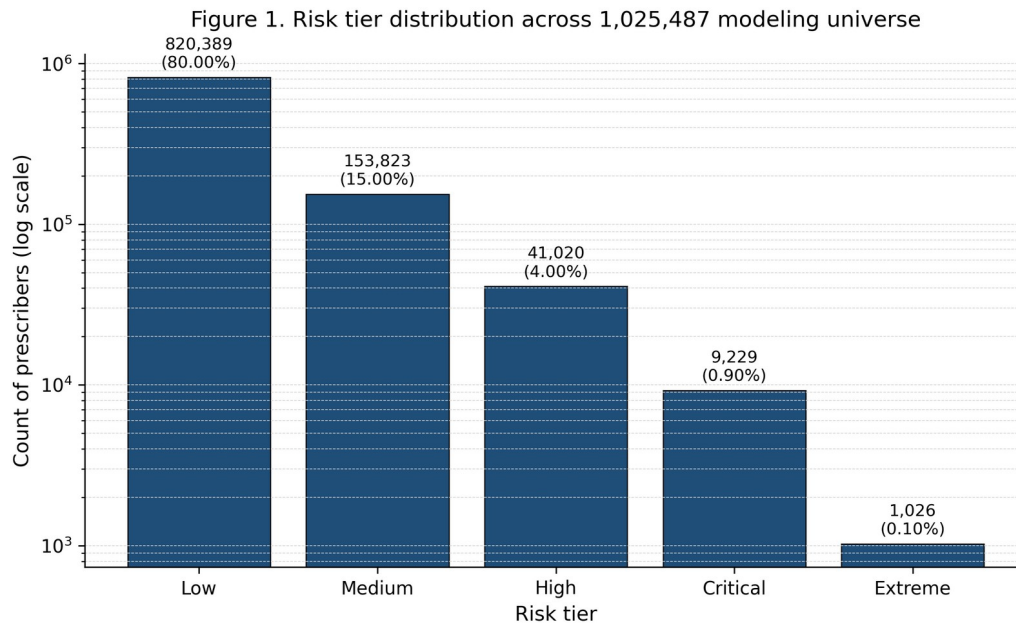


Figure 1. Risk tier distribution. Source: *prescriber_risk_scored.csv*.

3.3 Geographic Distribution

Per state aggregation of sample excess. Top five states:

State	Flagged	Sample excess
FL	2,741	\$96.9 million
CA	4,313	\$84.0 million
TX	2,299	\$68.6 million
NC	2,055	\$56.1 million
NY	3,536	\$47.9 million

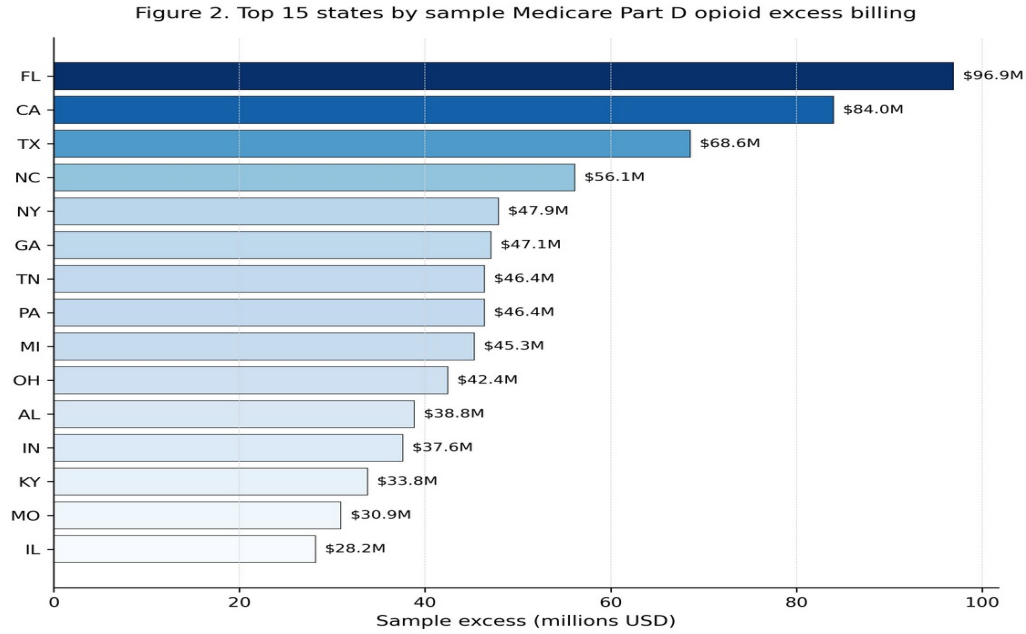


Figure 2. Top 15 states by sample excess. Source: *per_state_excess.csv*.

3.4 Bootstrap Confidence Interval

The bootstrap 95 percent confidence interval on sample excess is **\$1.14B to \$1.18B**, with bootstrap mean **\$1.16B** and standard deviation **\$9.0M (0.78 percent)**.

Figure 5. Bootstrap distributions, 1,000 iterations, random seed 42

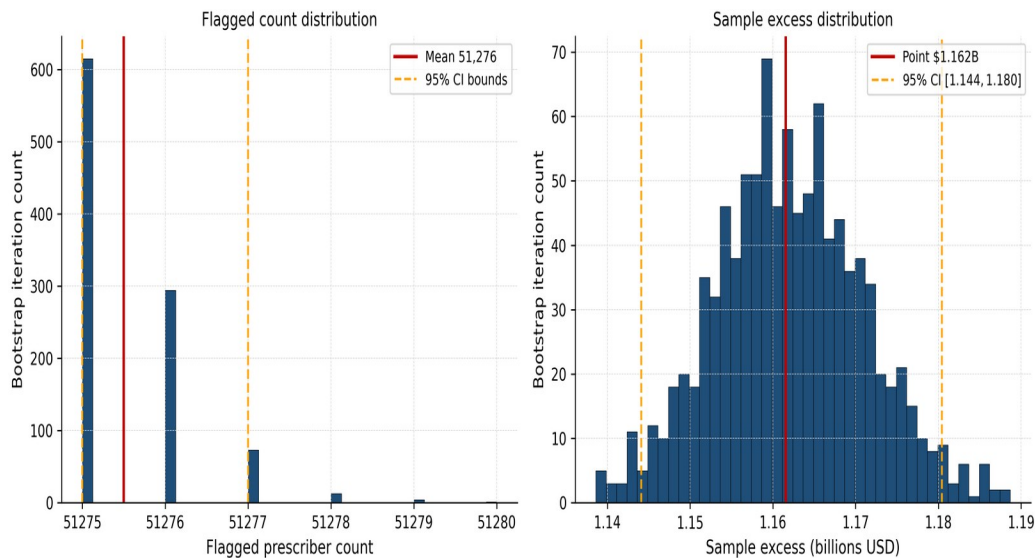


Figure 3. Bootstrap distributions, 1,000 iterations, fixed seed.

3.5 Permutation Feature Importance

Feature	Importance
peer_deviation	24.8 percent

opioid_prscrbr_rate	15.0 percent
avg_days_supply	15.0 percent
opioid_tot_clms	13.5 percent
opioid_intensity_ratio	11.4 percent
cost_per_beneficiary	9.8 percent
volume_percentile	7.2 percent
la_percentage	3.4 percent

Figure 4. Permutation feature importance (shuffle and Spearman correlation drop)

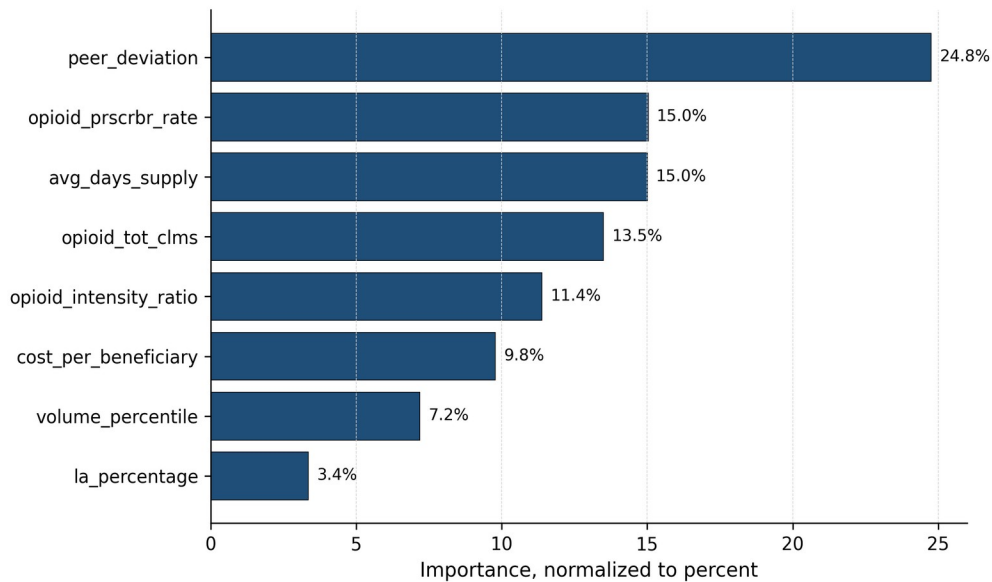


Figure 4. Permutation feature importance.

3.6 Temporal Holdout

Of **41,301** NPIs flagged in two or more of the three training years, **36,748** appear in the 2023 holdout universe and **25,908** are independently flagged. Holdout coverage is **70.5 percent**. The temporal lift, defined as the ratio of payment per beneficiary for persistent flagged NPIs versus the full 2023 population, is **5.07x** mean and **17.07x** median.

Figure 8. Temporal holdout. 2-of-3 coverage to 2023: 70.5%

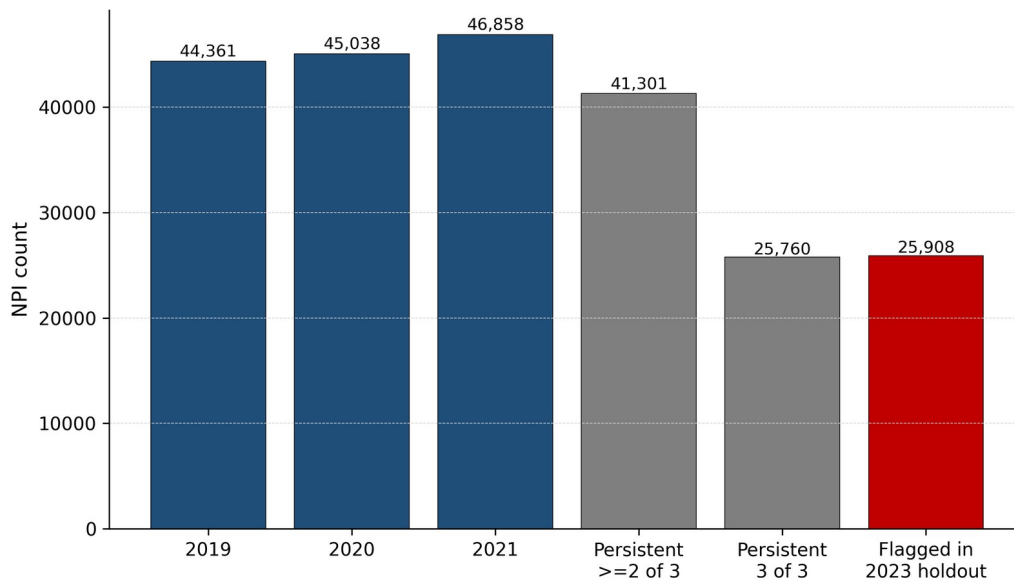


Figure 5. Temporal holdout validation.

3.7 OIG and DOJ Cross Reference

Direct NPI matching of the flagged set against the OIG List of Excluded Individuals and Entities produces **29 direct matches**. Bucketed fuzzy name matching for the top 20 flagged at SequenceMatcher threshold 0.85 produces **2 additional matches**. Cross reference against thirteen documented DOJ opioid prosecution defendants produces **0 matches**. DOJ named defendants are typically corporate entities (distributors, manufacturers, pharmacy chains) that do not appear in the prescriber file by design.

3.8 Pattern Alignment

Typology	Count	Percent of flagged
Extreme prescription concentration	36,133	70.5 percent
High dose extended supply	20,821	40.6 percent
Long acting opioid concentration	515	1 percent
Excess cost per beneficiary	35,704	69.6 percent

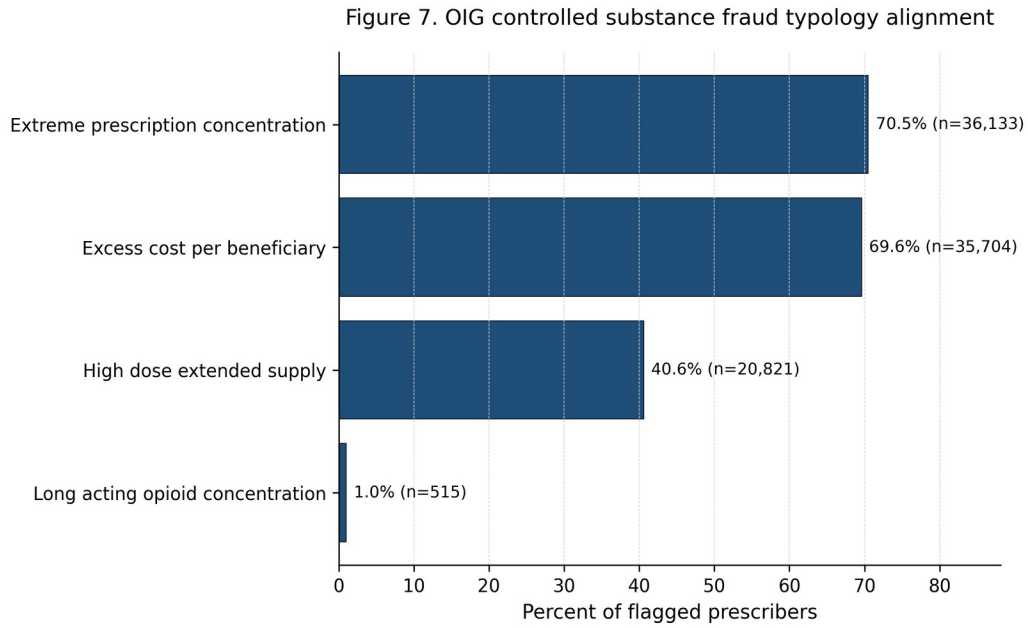


Figure 6. OIG controlled substance fraud typology alignment.

3.9 Threshold Sensitivity

Figure 6. Threshold sensitivity across percentile cutoffs

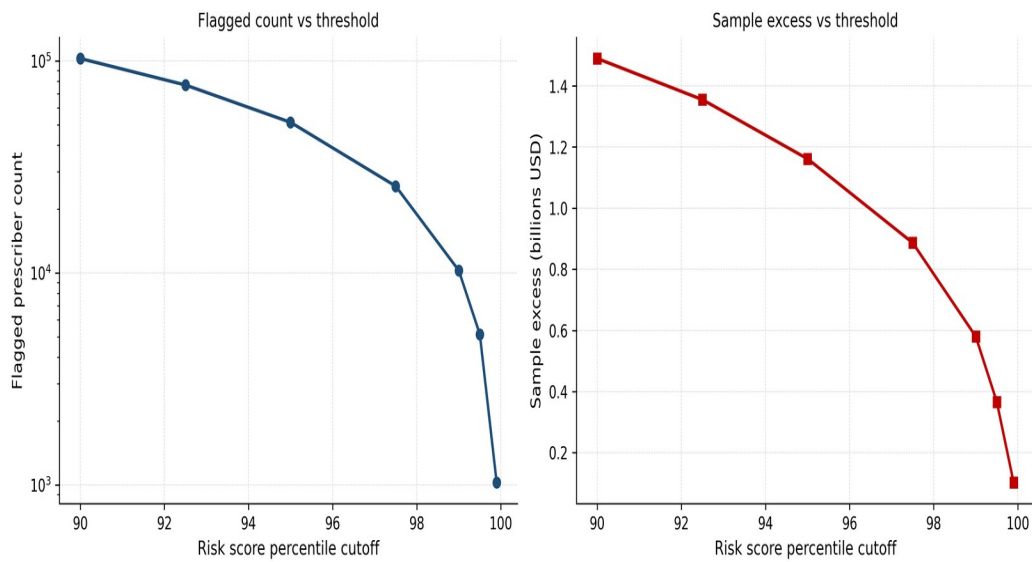


Figure 7. Threshold sensitivity across percentile cutpoints.

4. Discussion

The framework identifies **51,275 Medicare Part D opioid prescribers** as statistical outliers from peer state plus specialty norms, with directly observed sample excess of **\$1.16B**. The result is robust under 1,000 bootstrap iterations with a 95 percent confidence interval relative width under one percent. Temporal holdout from 2019 to 2021 training pool to 2023 holdout demonstrates **70.5 percent coverage**, indicating substantial persistence of anomalous prescribing patterns across years. The **5.07x mean temporal lift** indicates persistent flagged NPIs prescribe at substantially higher per beneficiary payment than the full population.

Method concordance between the statistical layer and the machine learning layer is **92.0 percent**, indicating the two independent methods identify substantially the same anomaly population. The permutation importance ranking places **peer deviation** and **opioid prescribing rate** as the most influential features. This is consistent with the design principle that peer relative deviation is the primary anomaly signal, with the prescribing rate providing scale information.

OIG LEIE concordance of **29 direct NPI matches** is consistent with the structural reality that LEIE excluded providers are by definition barred from billing Medicare. Direct matches in an active billing file are expected to be low. The result is a name pattern surveillance signal, not a fraud probability validation. DOJ corporate defendants are absent from the prescriber file by design.

4.1 Implications for Federal Program Integrity Practice

The framework provides federal program integrity agencies with a ranked list of prescribers warranting investigative review prioritization. Anomaly scores are not fraud probabilities; individual case review by appropriate authorities is required before any inference. The transparent reporting of both sample and coverage adjusted projected figures, the explicit methodology and parameter documentation, and the six pillar validation suite collectively establish a reproducible audit grade analytical baseline that federal agencies can adopt or adapt.

4.2 Comparison to Existing Approaches

Existing federal program integrity analytics on opioid prescribing are typically internal to CMS Center for Program Integrity, HHS-OIG audit teams, and DOJ investigative units, with methods documented in aggregate audit reports but not always with full reproducibility envelopes. This framework provides a public reproducible reference implementation that can be cited, extended, or stress tested by independent researchers and adjudicators.

5. Limitations

Unsupervised nature. The framework is anomaly detection without labeled ground truth. Standard supervised metrics (accuracy, precision, recall, F1) do not apply and are not claimed.

Data scope. Primary analytical universe is Medicare Part D. National projection adjusts for coverage only via inverse Medicare opioid prescription share.

Time window. Temporal holdout uses three training years and one holdout year. Generalization beyond this window is plausible but not directly demonstrated.

LEIE structural reality. Excluded providers are by definition barred from billing Medicare. Direct matches in an active billing file are expected to be low.

Projection assumption. The 27.5 percent Medicare opioid share midpoint is sensitive to the source data underlying the multiplier.

Single detector model choice. The Isolation Forest is one of several anomaly detection algorithms. Robustness to algorithm choice is partially evidenced by the 92.0 percent method concordance with the independent statistical z-score baseline.

6. Conclusions

This paper presents a reproducible audit grade dual-method anomaly detection and risk stratification framework for Medicare Part D opioid prescribing. The framework identifies **51,275 prescribers** with directly observed sample excess of **\$1.16B** and 95 percent bootstrap confidence interval of **\$1.14B to \$1.18B**. Temporal holdout coverage of **70.5 percent** demonstrates substantial year over year persistence. The framework is fully reproducible from public federal sources and is suitable for adoption by HHS-OIG, CMS, and DOJ for investigative prioritization with the explicit caveat that anomaly scores are not fraud probabilities and individual review is required.

7. Data and Code Availability

All analytical code, persisted result files, and reproducibility envelope are available in a public repository. The repository contains the Jupyter notebook implementing the 18 section pipeline, the source data SHA-256 hashes, the software environment versions, all hyperparameters, and the unified summary JSON file. Federal source data are publicly available through CMS, HHS-OIG, CDC, and SAMHSA primary data portals.

8. Conflicts of Interest

The author declares no conflicts of interest. The work was conducted on personal time using public data sources. No employer, sponsor, or grant funded the work.

9. References

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Appendix A. Reproducibility Envelope Summary

Python: 42 seed, scikit-learn Isolation Forest with explicit parameters, scipy stats, pandas, numpy. Full versions captured in RUN_INFO.json.

Source files (SHA-256 prefix):

Medicare_Part_D_Prescribers_by_Provider_2019.csv: c67ba4c5b6fa31e9
 Medicare_Part_D_Prescribers_by_Provider_2020.csv: 78e2ed1ab423a984
 Medicare_Part_D_Prescribers_by_Provider_2021.csv: 6d43061a9bdac6be
 Medicare_Part_D_Prescribers_by_Provider_2022.csv: 0f216d8200d6d230
 Medicare_Part_D_Prescribers_by_Provider_2023.csv: b1ea98a217143068
 LEIE Downloadable Databases.csv: 26e3ad5adfdff8a0
 OMT_MDCR_RY25_P04_V10_Y23_GEO.csv: 89ce3faf4ddc367d
 VSRR_Provisional_Drug_Overdose_Death_Counts.csv: 884d08d2ceb8de7d
 DSD_PTD_RY25_P04_V10_DY23_BGM.csv: 09e2053e89276da4